

Study Guide 5: Duration, Average Life and Convexity

Fixed Income Study Guides

These three measures describe when a bond's money comes back and how its price reacts to rate changes. Duration is the first-order rate sensitivity, convexity corrects for the curve in the price-yield relationship, and average life measures only when principal is repaid.

Worked bond (from Study Guide 4): 10-year, 5% semiannual coupon, price 95, YTM 5.6617%. Results: Macaulay duration **7.927**, modified duration **7.709**, convexity **72.41**, DV01 **0.0732** per 100 face.

1. Macaulay duration

The weighted-average time to receive the bond's cash flows, where each weight is that cash flow's share of the price. Measured in years.

$$D_{mac} = \sum_t t \times w_t \quad w_t = \frac{CF_t / (1 + y)^t}{P}$$

For semiannual bonds, compute with t in half-years and y per half-year, then divide the result by 2.

Closed form at a coupon date, with y and c per period and T periods:

$$D_{mac} = \frac{1 + y}{y} - \frac{(1 + y) + T(c - y)}{c[(1 + y)^T - 1] + y}$$

Worked bond: $y = 2.8308\%$, $c = 2.5\%$, $T = 20$, giving 15.855 half-years = **7.927 years**.

Check against the book: an 8% coupon, 2-year bond at a 10% yield has duration 1.8852 years; a 2-year zero has duration 2.0.

2. Modified duration

The percentage price change for a 1-unit change in yield. This is the number used for risk.

$$D^* = \frac{D_{mac}}{1 + y/k} \quad \frac{\Delta P}{P} \approx -D^* \Delta y$$

Here k is payments per year. Worked bond: $7.927/1.028308 = 7.709$. A 100 bp rise predicts a 7.71% price fall.

3. Rules for duration

Rule	Statement	Why
1	A zero's duration equals its maturity	All value arrives at one date
2	Holding maturity fixed, lower coupon means higher duration	More value sits in the final payment

Rule	Statement	Why
3	Holding coupon fixed, duration generally rises with maturity	Cash flows are further out; deep-discount long bonds are the exception
4	Holding coupon and maturity fixed, lower yield means higher duration	Distant cash flows are discounted less, so they weigh more
5	A level perpetuity has duration $(1 + y) \div y$	At 10%, duration = 11 years

$$D_{perpetuity} = \frac{1 + y}{y}$$

4. Dollar duration and DV01

Traders manage dollars, not percentages.

$$\text{Dollar duration} = D^* \times P \quad DV01 = D^* \times P \times 0.0001$$

- Worked bond: $DV01 = 7.709 \times 95 \times 0.0001 = 0.0732$ per 100 face. For \$10 million face, a 1 bp rise costs about **\$7,324**.
- DV01 is also called PVBP (price value of a basis point) or BPV. DV01s add across positions, which makes them the standard unit for limits and hedges.

5. Effective duration

For bonds whose cash flows change when rates change (callables, puttables, MBS), YTM-based duration is wrong. Reprice with a model at yields shifted up and down:

$$D_{eff} = \frac{P_- - P_+}{2 P_0 \Delta y}$$

Worked bond with $\Delta y = 50$ bp: $P_- = 98.749$, $P_+ = 91.423$, so

$$D_{eff} = \frac{98.749 - 91.423}{2 \times 95 \times 0.005} = 7.712$$

matching modified duration for an option-free bond. For a callable, P_- is capped near the call price, so effective duration falls well below modified duration.

6. Portfolio and key rate duration

- **Portfolio duration** is the value-weighted average of the holdings' durations. A strictly correct figure uses one common yield, but the weighted average is standard practice.
- **Key rate durations** measure sensitivity to one point on the curve (2, 5, 10, 30 years) at a time. They capture twists and steepening that a single duration, which assumes a parallel shift, misses.

$$D_{portfolio} = \sum_i w_i D_i \quad w_i = \frac{MV_i}{\sum_j MV_j}$$

7. Weighted average life (WAL)

The average time until each dollar of **principal** is repaid. It ignores interest and discounting, so it measures timing of principal, not price risk.

$$WAL = \frac{\sum_t t \times \text{Principal}_t}{\sum_t \text{Principal}_t}$$

Instrument	WAL	Note
Worked bond (bullet)	10.00 yr	All principal at maturity; compare duration 7.93
Bond with sinking fund retiring 20% a year, years 1-5	3.00 yr	$(1 + 2 + 3 + 4 + 5) \times 20 \div 100$
30-year 6% level-payment mortgage, no prepayment	19.31 yr	Principal repaid slowly at first
Same mortgage, 8% CPR prepayments	8.96 yr	Prepayments cut WAL by more than half
15-year 6% level-payment mortgage	8.65 yr	

Uses:

- **MBS and ABS pricing:** spreads are quoted over the Treasury whose maturity is nearest the security's WAL.
- **Amortizing loans and sinking-fund bonds:** WAL gives a single "effective maturity" for comparing issues.
- **Bank asset-liability management:** WAL shows when principal returns for reinvestment.
- **Blind spot:** WAL depends on a prepayment assumption and says nothing about price sensitivity. Use effective duration for risk.

8. Convexity

The price-yield curve is convex, so duration alone underestimates price gains and overstates price losses. Convexity measures that curvature.

$$C = \frac{1}{P(1+y)^2} \sum_t \frac{CF_t}{(1+y)^t} (t^2 + t)$$

With t in half-years and y per half-year, divide the result by 4 to annualize. Worked bond: **C = 72.41**. For a zero:

$$C_{zero} = \frac{T(T+0.5)}{(1+y/2)^2}$$

A 30-year strip at 4.5% has $C = 875$.

Price change with convexity:

$$\frac{\Delta P}{P} \approx -D^* \Delta y + \frac{1}{2} C (\Delta y)^2$$

Yield change	Actual price change	Duration only	Duration + convexity
+200 bp	-14.06%	-15.42%	-13.97%
+100 bp	-7.36%	-7.71%	-7.35%
-100 bp	+8.08%	+7.71%	+8.07%
-200 bp	+16.97%	+15.42%	+16.87%

Duration alone is off by 1.4-1.6 points at ± 200 bp; adding convexity cuts the error to about 0.1.

Effective convexity for bonds with options:

$$C_{eff} = \frac{P_- + P_+ - 2P_0}{P_0 (\Delta y)^2}$$

Worked bond at ± 50 bp:

$$C_{eff} = \frac{98.749 + 91.423 - 190}{95 \times 0.005^2} = 72.4$$

Why investors like convexity: for two bonds with equal duration, the more convex one gains more when rates fall and loses less when rates rise. The market charges for it with a lower yield.

Negative convexity: callable bonds and mortgage-backed securities. As rates fall, the call or prepayments cap the price near par, so the price curve bends the wrong way. The holder gets small gains and full-sized losses. This is why MBS yield more than Treasuries of similar duration.

9. Using the measures to manage interest rate risk

Immunization protects a known future liability against rate changes. Conditions:

1. PV of assets = PV of liability.
2. Duration of assets = duration of liability.
3. Convexity of assets > convexity of liability (so any rate move leaves a surplus).

Example: a \$1,000,000 liability due in 7 years, with rates flat at 10% (annual). PV = \$513,158; duration = 7. Fund it with a 3-year zero (duration 3) and a perpetuity (duration 11):

$$w \times 3 + (1 - w) \times 11 = 7 \Rightarrow w = 0.5$$

Buy \$256,579 of each. Durations drift at different speeds as time passes and rates move, so the portfolio must be **rebalanced**. Buying a 7-year zero (cash flow matching, dedication) avoids rebalancing entirely; that is the strip use case from Study Guide 1.

DV01 hedging. The hedge ratio is:

$$\text{Hedge ratio} = \frac{DV01_{\text{position}}}{DV01_{\text{hedge}}}$$

- Position: \$10 million face of the worked bond, DV01 \$7,324.
- Hedge with a 30-year strip at 4.5%: $DV01 = 29.34 \times 26.315 \times 0.0001 = 0.0772$ per 100.
- Short about **\$9.49 million face** of the strip (market value about \$2.50 million).
- The hedge is exact only for a parallel shift. It leaves curve risk (10-year vs 30-year) and a convexity mismatch (72 vs 875). Key rate DV01s or a Treasury futures hedge in the 10-year sector reduce that residual risk.

Other uses:

- **Bank and insurer gap management:** match asset and liability durations to protect net worth, not just income.
- **Portfolio positioning:** active managers lengthen duration when they expect falling rates and shorten it when they expect rising rates.
- **Risk limits and reporting:** DV01 and key rate DV01 limits by desk, with convexity reported for large moves.

10. Tools: the worked bond on every platform

Targets: Macaulay 7.927 · modified 7.709 · convexity 72.41 · DV01 0.0732 · prices at ±50 bp 98.749 / 91.423 · mortgage WAL 19.31.

TI BA II Plus Professional

Modified and Macaulay duration (Bond worksheet)

1. 2nd BOND: SDT = 11.1526, CPN = 5, RDT = 11.1536, RV = 100, ACT, 2/Y.
2. PRI = 95 ENTER, then YLD CPT → 5.6617.
3. ↓ to DUR → **7.709** (modified).
4. Macaulay: $7.709 \times (1 + 5.6617 \div 200) = \rightarrow \mathbf{7.927}$.

Effective duration and convexity (TVM, P/Y = 2)

1. 20 N · 2.5 PMT · 100 FV.
2. 5.1617 I/Y, CPT PV → -98.749; STO 1 (after +|-).
3. 6.1617 I/Y, CPT PV → -91.423; STO 2 (after +|-).
4. Duration: $(\text{RCL } 1 - \text{RCL } 2) \div (2 \times 95 \times .005) = \rightarrow \mathbf{7.712}$.
5. Convexity: $(\text{RCL } 1 + \text{RCL } 2 - 190) \div (95 \times .005 \times^2) = \rightarrow \mathbf{72.4}$.

DV01: $7.709 \times 95 \times .0001 = \rightarrow 0.0732$.

WAL of amortizing loans: the AMORT worksheet gives principal paid over any range of payments (P1 to P2). Group payments by year and weight by year for an approximate WAL; use Excel for exact monthly results.

HP 12C

The 12C has no duration or convexity keys, so use repricing.

1. f CLEAR FIN; 20 n · 2.5 PMT · 100 FV.
2. 5.1617 ENTER 2 ÷ i, PV → -98.749; CHS STO 1.
3. 6.1617 ENTER 2 ÷ i, PV → -91.423; CHS STO 2.
4. Effective duration: $\text{RCL } 1 \text{ RCL } 2 - .95 \div \rightarrow \mathbf{7.712}$ (dividing by $2 \times 95 \times 0.005 = 0.95$).
5. Effective convexity: $\text{RCL } 1 \text{ RCL } 2 + 190 - .002375 \div \rightarrow \mathbf{72.4}$ ($95 \times 0.005^2 = 0.002375$).
6. Modified to Macaulay: 7.709 ENTER $1.028308 \times \rightarrow 7.927$.

WAL: f AMORT after n payments shows interest, then x↵y shows principal paid, year by year (12 f AMORT repeatedly).

Excel

Measure	Formula	Result
Macaulay duration	=DURATION(DATE(2026,11,15),DATE(2036,11,15),5%,5.6617%,2,1)	7.927
Modified duration	=MDURATION(DATE(2026,11,15),DATE(2036,11,15),5%,5.6617%,2,1)	7.709
Price at -50 bp	=PRICE(DATE(2026,11,15),DATE(2036,11,15),5%,5.1617%,100,2,1)	98.749
Effective duration	=(P_minus - P_plus)/(2*95*0.005)	7.712

Measure	Formula	Result
Effective convexity	$=(P_{\text{minus}}+P_{\text{plus}}-2*95)/(95*0.005^2)$	72.4
DV01 per 100	$=MDURATION(...)*95*0.0001$	0.0732
Mortgage WAL (Excel 365)	$=SUMPRODUCT(SEQUENCE(360)/12*PMT(6\%/12, SEQUENCE(360), 360, -100000))/100000$	19.31

Excel has no convexity function. Build a cash flow table (periods in A2:A21, cash flows in B2:B21, yield in E1, price in E2):

$$=SUMPRODUCT(B2:B21/(1+E1/2)^{A2:A21}*A2:A21*(A2:A21+1))/(E2*(1+E1/2)^2)/4$$

This mirrors the book's Spreadsheet 16.1, which builds duration cash flow by cash flow.

Python

QuantLib (tested; output 7.927, 7.709, 72.41, -0.0732):

```
import QuantLib as ql
settle = ql.Date(15, ql.November, 2026)
ql.Settings.instance().evaluationDate = settle
sched = ql.Schedule(settle, ql.Date(15, ql.November, 2036), ql.Period(ql.Semiannual),
                    ql.NullCalendar(), ql.Unadjusted, ql.Unadjusted,
                    ql.DateGeneration.Backward, False)
dc = ql.ActualActual(ql.ActualActual.Bond, sched)
bond = ql.FixedRateBond(0, 100.0, sched, [0.05], dc)
ytm = bond.bondYield(ql.BondPrice(95.0, ql.BondPrice.Clean), dc, ql.Compounded, ql.Semiannual)
rate = ql.InterestRate(ytm, dc, ql.Compounded, ql.Semiannual)
mac = ql.BondFunctions.duration(bond, rate, ql.Duration.Macaulay)
mod = ql.BondFunctions.duration(bond, rate, ql.Duration.Modified)
conv = ql.BondFunctions.convexity(bond, rate)
dv01 = ql.BondFunctions.basisPointValue(bond, rate) # sign: price change for +1 bp
```

Plain NumPy, plus mortgage WAL with numpy-financial (tested):

```
import numpy as np, numpy_financial as npf
y, t = 0.0566169, np.arange(1, 21)
cf = np.full(20, 2.5); cf[-1] += 100
pv = cf / (1 + y / 2) ** t; P = pv.sum()
mac = (t * pv).sum() / P / 2 # 7.927
mod = mac / (1 + y / 2) # 7.709
conv = (pv * t * (t + 1)).sum() / (P * (1 + y / 2) ** 2) / 4 # 72.41

k = np.arange(1, 361)
prin = npf.pmt(0.06 / 12, k, 360, -100_000)
wal = (k / 12 * prin).sum() / prin.sum() # 19.31
```

Julia

Plain Julia (same logic as the tested NumPy code; not run here):

```
y = 0.0566169; t = 1:20
cf = fill(2.5, 20); cf[end] += 100
pv = cf ./ (1 + y / 2) .^ t; P = sum(pv)
mac = sum(t .* pv) / P / 2
modd = mac / (1 + y / 2)
conv = sum(pv .* t .* (t .+ 1)) / (P * (1 + y / 2)^2) / 4
```

mod is a built-in Julia function, hence modd. ActuaryUtilities.jl provides duration (Macaulay, modified, key rate) and convexity functions that take a yield or a fitted FinanceModels.jl curve.

11. Practice set

1. A 5-year 6% semiannual bond trades at par. Find Macaulay and modified duration and convexity. 4.393, 4.265, 21.77.
2. Duration of a perpetuity at 8%? *Answer:* $1.08/0.08 = 13.5$ years.
3. Estimate the worked bond's price change for +50 bp with duration and convexity. Compare with the actual change. *Answer:* $-7.709 \times 0.005 + \frac{1}{2} \times 72.41 \times 0.005^2 = -3.764\%$; actual -3.765% .
4. Immunize a 5-year liability using 2-year and 10-year zeros. Weights? *Answer:* $w \times 2 + (1 - w) \times 10 = 5$, so 62.5% in the 2-year and 37.5% in the 10-year.
5. A 10-year bond retires 25% of principal in each of years 7-10. WAL? $(7 + 8 + 9 + 10) \div 4 = 8.5$ years.
6. Why does a 30-year strip need more rebalancing in a DV01 hedge than a 10-year note? *Its convexity (875 vs 72) makes its DV01 change much faster as rates move.*
7. Why do MBS show negative convexity? *Falling rates speed up prepayments, returning principal at par just when the bond would rise above it.*

12. CFA-style review questions

Original questions in the CFA Level I format (three choices, one correct). Not CFA Institute material.

1. A bond has modified duration 7.0 and convexity 60. For a 100 bp rise in yield, the estimated price change is closest to:
 - A. -7.30%
 - B. -7.00%
 - C. -6.70%
2. Which bond has the highest duration?
 - A. a 10-year 8% coupon bond
 - B. a 10-year zero-coupon bond
 - C. a 5-year zero-coupon bond
3. The Macaulay duration of a level perpetuity yielding 5% is closest to:
 - A. 20 years
 - B. 21 years
 - C. 22 years
4. A bond priced at 100 falls to 98.20 when yields rise 50 bp and rises to 102.00 when yields fall 50 bp. Its effective duration is closest to:
 - A. 1.9
 - B. 3.8
 - C. 7.6
5. A \$5 million face position has modified duration 6.0 and a price of 98. Its DV01 is closest to:
 - A. \$2,940
 - B. \$3,000
 - C. \$29,400
6. When interest rates fall sharply, a callable bond's price is most likely to:
 - A. rise more than an otherwise identical option-free bond
 - B. rise less than an otherwise identical option-free bond
 - C. fall, because the call is exercised

Answer key

1. C. $-7.0 \times 0.01 + \frac{1}{2} \times 60 \times 0.01^2 = -7.00\% + 0.30\% = -6.70\%$.

2. B. A zero's duration equals its maturity, the longest here.
3. B. $1.05/0.05 = 21$.
4. B. $(102.00 - 98.20)/(2 \times 100 \times 0.005) = 3.8$.
5. A. $6.0 \times 98 \times 0.0001 = 0.0588$ per 100; $\times 50,000 = \$2,940$.
6. B. Negative convexity: the call caps price appreciation near the call price.

13. Bloomberg Terminal exercises (draft: verify on the campus Terminal)

Command	What to do	Check against this guide	Verified
CT10 Govt YAS <G0>	Read modified duration, convexity and risk (DV01 per 100)	Sections 2, 4 and 8	☐
YAS <G0>, hedge area	Set a hedge instrument (for example the current 30-year or a Treasury future) and read the hedge ratio	Section 9: DV01 hedging	☐
OAS1 <G0>	Open a callable bond; compare option-adjusted (effective) duration and convexity with the yield-based figures	Sections 5 and 8: effective duration, negative convexity	☐
KRR <G0> (confirm)	Key rate risk for a bond or portfolio	Section 6: key rate duration	☐
PRTU <G0>, then PORT <G0>	Build a small test portfolio (say 2-, 10- and 30-year Treasuries); read portfolio duration and key rate exposures	Section 6: portfolio duration	☐
DES <G0> on an agency MBS pool	Read weighted average life and prepayment speed (CPR); change the prepayment assumption and watch WAL	Section 7: WAL	☐

Excel (confirm with FLDS): `=BDP("CT10 Govt", "DUR_ADJ_MID")`, `=BDP("CT10 Govt", "CONVEXITY_MID")`, `=BDP("CT10 Govt", "RISK_MID")`.

Exercise: for \$10 million face of the current 10-year, compute DV01 from RISK_MID. Size a hedge in the current 30-year using the ratio of their DV01s, then compare your answer with the YAS hedge area.

14. Reading in Bodie, Kane & Marcus, *Investments*, 13th ed. (2024)

Section	Topic in the book	Supports
16.1 Interest Rate Risk	Interest Rate Sensitivity; Duration; What Determines Duration? (Rules 1-5); Spreadsheet 16.1	Sections 1-4 and the duration rules
16.2 Convexity	Why Do Investors Like Convexity?; Duration and Convexity of Callable Bonds; Duration and Convexity of Mortgage-Backed Securities	Sections 5 and 8: convexity, effective duration, negative convexity
16.3 Passive Bond Management	Bond-Index Funds; Immunization; Cash Flow Matching and Dedication; Other Problems with Conventional Immunization	Section 9: immunization and rebalancing
16.4 Active Bond Management	Sources of Potential Profit; Horizon Analysis	Duration positioning
14.5 Default Risk and Bond Pricing	Bond Indentures: Sinking Funds	WAL of sinking-fund bonds
2.2 The Bond Market	Mortgage and Asset-Backed Securities	WAL and prepayment background
20.5 Option-like Securities	Callable Bonds	Why callables have negative convexity
23.3 Interest Rate Futures	Hedging Interest Rate Risk	DV01 hedging with futures
23.4 Swaps	Swaps and Balance Sheet Restructuring	Changing portfolio duration with swaps

Chapter 16's end-of-chapter problems and CFA problems give extensive duration, convexity and immunization practice. Weighted average life is not a named topic in the book; it comes up through sinking funds and mortgage-backed securities.